

DNS — Short Revision Notes

1. What is DNS?

DNS (Domain Name System) resolves a **human-readable name** into an **IP address**.

google.com → IP address

- Humans use **names**.
- Computers communicate using **IPv4/IPv6 addresses**.
- DNS acts like the **phonebook of the network**.

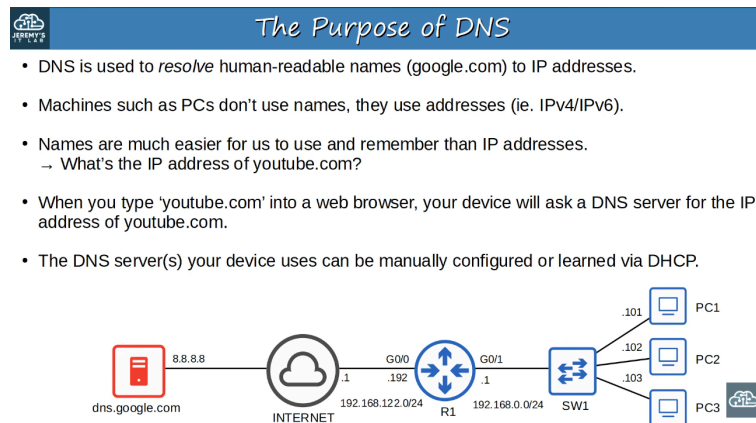
Why DNS?

Instead of remembering an IP address:

172.x.x.x

we can use:

[youtube.com](https://www.youtube.com)

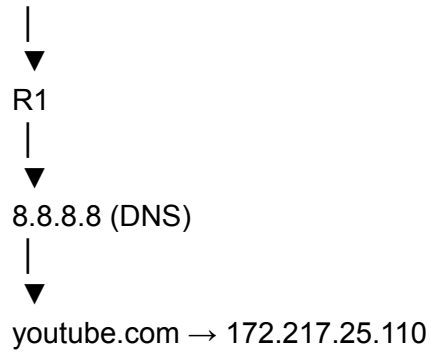


2. Basic DNS Flow

PC → DNS Server → IP Address → Destination

Example:

PC1



Important: The router does not have to be a DNS server. It can simply **forward DNS packets**.

3. DNS Server Configuration on a PC

Use:

```
ipconfig /all
```

It shows:

IPv4 Address
Subnet Mask
Default Gateway
DNS Servers

Example:

IPv4 Address : 192.168.0.101
Default Gateway : 192.168.0.1
DNS Server : 8.8.8.8

The DNS server can be:

- Manually configured
 - Learned through **DHCP**
-

4. nslookup

nslookup is used to query/test DNS.

nslookup youtube.com

Example:

Server: dns.google

Address: 8.8.8.8

Name: youtube.com

Address: 172.217.25.110

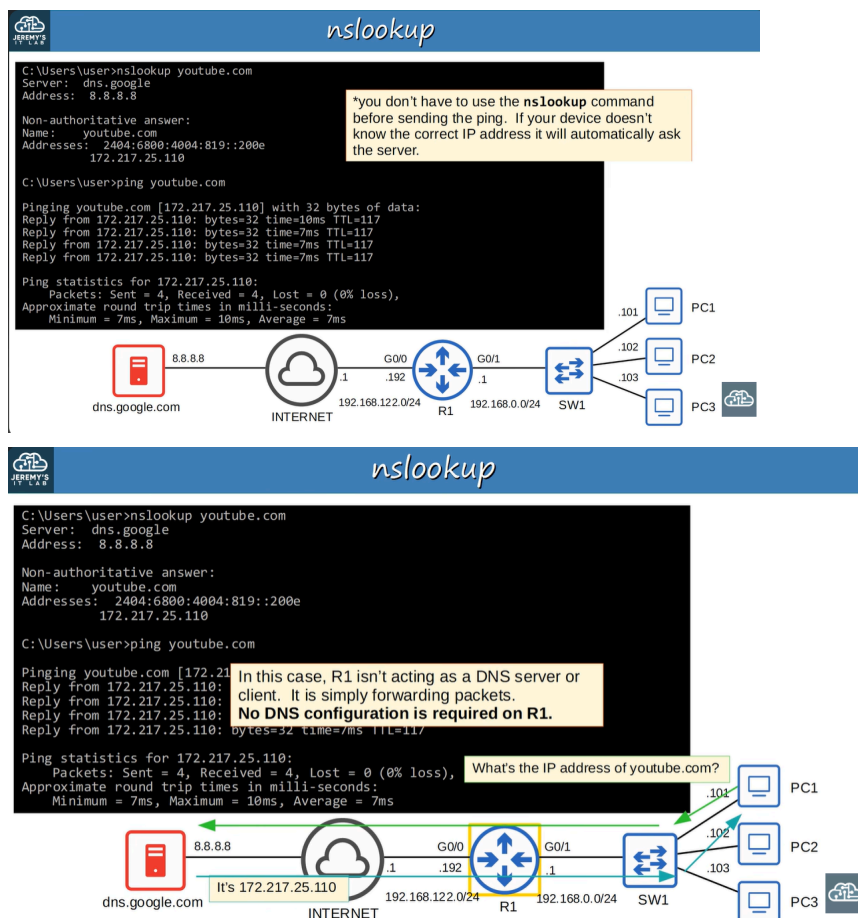
Meaning:

PC → Ask DNS server

DNS server → Return IP

Remember

nslookup = DNS testing/query tool



5. DNS Records

Record	Purpose
--------	---------

A	Name → IPv4
----------	-------------

AAAA	Name → IPv6
-------------	-------------

CNAME	Alias → another hostname
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Example:

youtube.com

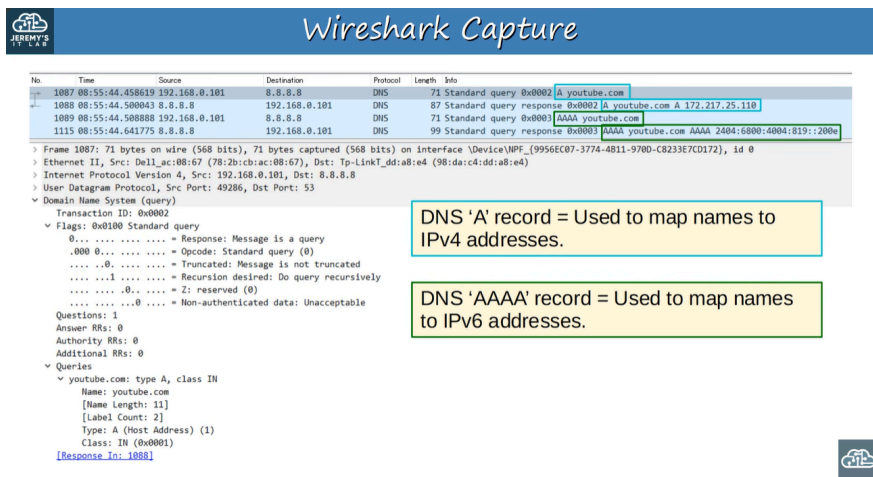
↓ A

172.217.25.110

youtube.com

↓ AAAA

2404:6800:4004:819::200e



The image shows a Wireshark network capture. The packet list at the top shows four DNS packets: a query for 'A' records of 'youtube.com', a response with IP 172.217.25.110, a query for 'AAAA' records of 'youtube.com', and a response with IPv6 address 2404:6800:4004:819::200e. The details pane on the left shows the structure of the first query packet, including flags, questions, and queries. Two callout boxes on the right explain the purpose of 'A' and 'AAAA' records.

No.	Time	Source	Destination	Protocol	Length	Info
1087	08:55:44.458619	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0002 A youtube.com
1088	08:55:44.500043	8.8.8.8	192.168.0.101	DNS	87	Standard query response 0x0002 A youtube.com A 172.217.25.110
1089	08:55:44.508888	192.168.0.101	8.8.8.8	DNS	71	Standard query 0x0002 AAAA youtube.com
1115	08:55:44.641775	8.8.8.8	192.168.0.101	DNS	99	Standard query response 0x0002 AAAA youtube.com AAAA 2404:6800:4004:819::200e

Transaction ID: 0x0002

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

- youtube.com: type A, class IN
- Name: youtube.com
- [Name Length: 11]
- [Label Count: 2]
- Type: A (Host Address) (1)
- Class: IN (0x0001)

[Response In: 1088]

DNS 'A' record = Used to map names to IPv4 addresses.

DNS 'AAAA' record = Used to map names to IPv6 addresses.

6. DNS Cache

Devices store DNS responses in a **local DNS cache**.

Why?

So they don't need to ask the DNS server every time.

First request:

PC → DNS Server → IP

Next request:

PC → Local DNS Cache → IP

Windows commands

View cache:

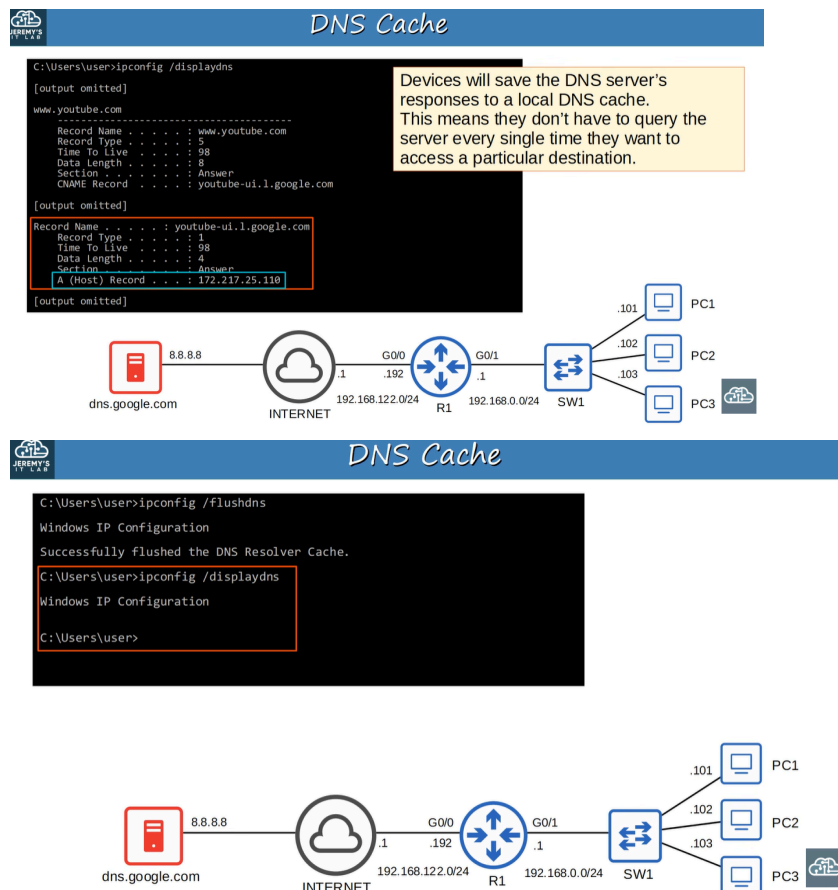
`ipconfig /displaydns`

Clear cache:

`ipconfig /flushdns`

Important

TTL (Time To Live) determines how long a DNS record can remain cached.



7. What Happens When You ping youtube.com?

You don't necessarily need to run `nslookup` first.

ping youtube.com

The PC first needs the IP address.

PC
↓
DNS lookup
↓
DNS Server
↓
IP address
↓
Ping destination

Example:

youtube.com → 172.217.25.110

Then:

ICMP → 172.217.25.110

Important

The router may simply **forward both DNS and ICMP packets**.

It doesn't have to perform DNS itself.

8. DNS in Cisco IOS

A Cisco router can work as:

DNS Client

The router asks a DNS server to resolve names.

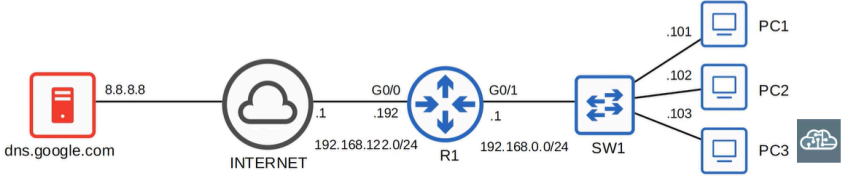
DNS Server

The router itself provides basic DNS resolution.



DNS in Cisco IOS

- For hosts in a network to use DNS, you don't need to configure DNS on the routers. They will simply forward the DNS messages like any other packets.
- However, a Cisco router can be configured as a DNS server, although it's rare.
→ If an internal DNS server is used, usually it's a Windows or Linux server.
- A Cisco router can also be configured as a DNS client.



9. Router as DNS Server

Enable DNS server:

```
R1(config)# ip dns server
```

Create hostname/IP mappings:

```
R1(config)# ip host R1 192.168.0.1
R1(config)# ip host PC1 192.168.0.101
R1(config)# ip host PC2 192.168.0.102
R1(config)# ip host PC3 192.168.0.103
```

Now the router knows:

```
PC1 → 192.168.0.101
PC2 → 192.168.0.102
PC3 → 192.168.0.103
```

Check the entries:

R1# show hosts

Remember

ip dns server

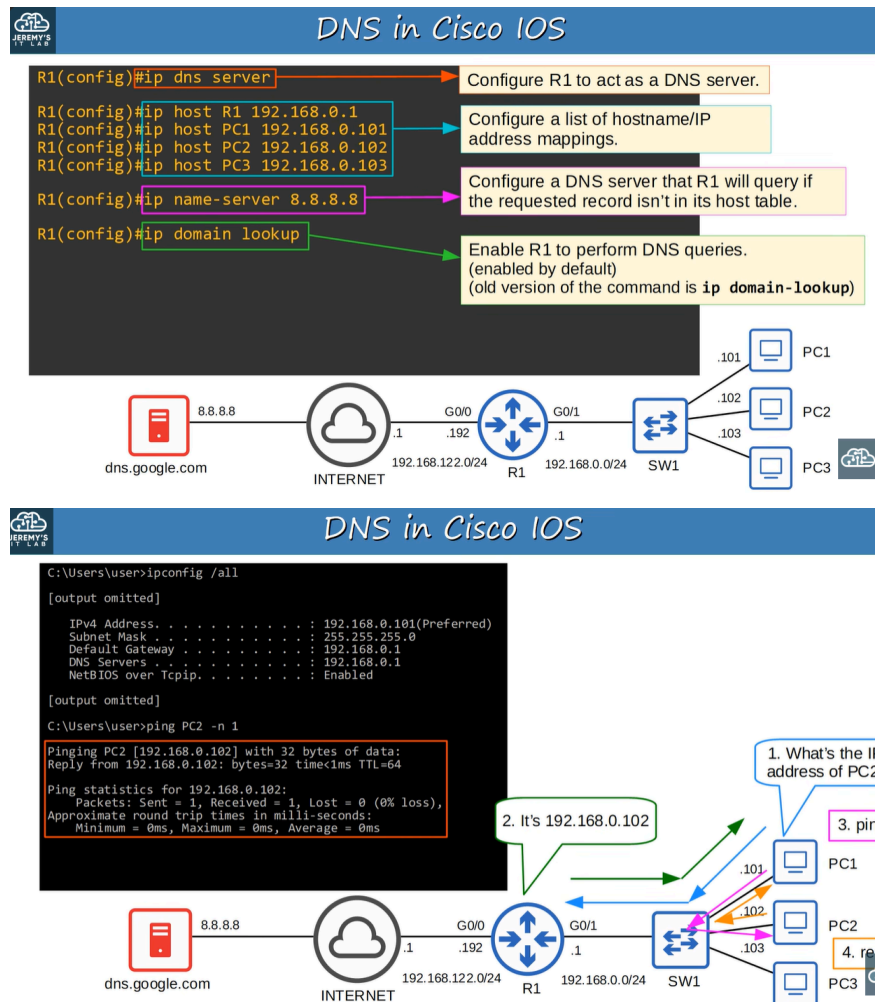


Router becomes DNS server

ip host NAME IP



Create local name → IP mapping



10. Router as DNS Client

Tell R1 which DNS server to use:

```
R1(config)# ip name-server 8.8.8.8
```

Now R1 can ask 8.8.8.8 to resolve names.

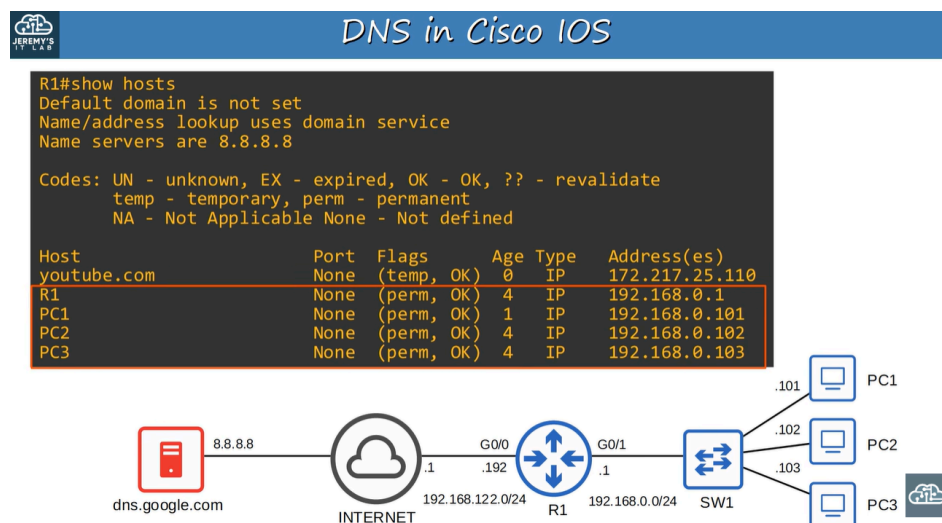
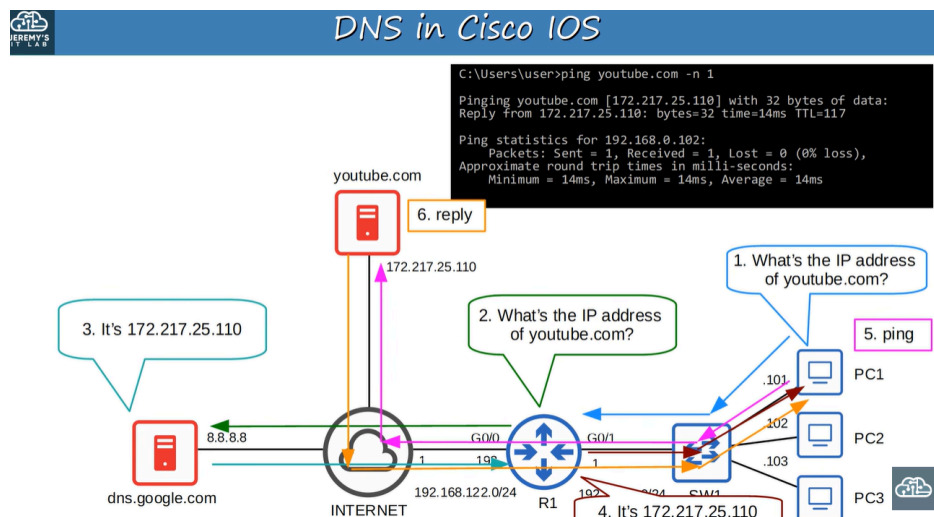
Enable DNS lookup:

```
R1(config)# ip domain lookup
```

Then:

```
R1# ping youtube.com
```

R1 performs DNS resolution and then sends the ICMP packets.



11. ip domain name

R1(config)# ip domain name example.com

Sets the router's **default domain name**.

For example:

ping pc1

can be interpreted as:

ping pc1.example.com

Remember

ip domain name

→ Sets default domain name



DNS in Cisco IOS

```
R1(config)#do ping youtube.com
Translating "youtube.com"
% Unrecognized host or address, or protocol not running.

R1(config)#ip name-server 8.8.8.8
R1(config)#ip domain lookup
R1(config)#do ping youtube.com
Translating "youtube.com"...domain server (8.8.8.8) [OK]

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.217.25.110, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/13 ms

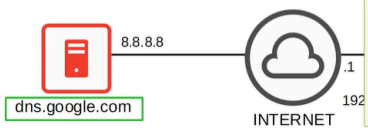
R1(config)#ip domain name jeremysitlab.com
```

Configure R1 to use the specified DNS server.

Enable R1 to perform DNS queries. (default)

(optional)

- Configure the default *domain* name.
- This will be automatically appended to any hostnames without a specified *domain*.
- ie. `ping pc1` will become `ping pc1.jeremysitlab.com`
- (old version of the command: `ip domain-name`)
- I will cover this command again in a later video (about SSH).



12. Cisco DNS Commands — Quick Table

Command	Purpose
<code>ip dns server</code>	Make router a DNS server
<code>ip host NAME IP</code>	Create local hostname → IP mapping

<code>ip name-server IP</code>	Configure DNS server for the router
<code>ip domain lookup</code>	Enable DNS lookup
<code>ip domain name NAME</code>	Set default domain name
<code>show hosts</code>	Show host/DNS entries

13. Router Does NOT Need DNS Configuration for PCs

Suppose:

PC1 → SW1 → R1 → Internet → DNS Server

PC1 has:

DNS Server = 8.8.8.8

R1 does **not** need:

`ip name-server 8.8.8.8`

just to forward PC1's DNS traffic.

Why?

Because R1 is simply routing the packet:

PC1
↓
DNS query
↓
R1
↓
8.8.8.8

★ Important

Routing decides how to reach the DNS server.

DNS decides what IP address belongs to a name.

14. Hosts File

The **hosts file** is a local file that manually maps:

Hostname → IP address

Example:

```
192.168.0.10  server1
192.168.0.20  server2
```

Now:

ping server1

can resolve:

server1 → 192.168.0.10

without needing DNS for that entry.

Windows location

C:\Windows\System32\drivers\etc\hosts

Remember

Hosts file → Local/static name resolution

DNS → Network/distributed name resolution



Quick Revision

DNS → Domain name → IP address

A → IPv4

AAAA → IPv6

CNAME → Alias

DNS → Port 53
UDP 53 → Mainly used
TCP 53 → Also used when required

nslookup → Test/query DNS

ipconfig /all
→ Shows DNS server

ipconfig /displaydns
→ Shows DNS cache

ipconfig /flushdns
→ Clears DNS cache

ip name-server 8.8.8.8
→ Router uses 8.8.8.8 as DNS server

ip dns server
→ Router becomes DNS server

ip host PC1 192.168.0.101
→ Local hostname/IP mapping

show hosts
→ Show router's host table

ip domain lookup
→ Enable DNS lookup

ip domain name example.com
→ Set default domain



Revision Comments & Doubts

Doubt — Does R1 need to be a DNS server?

Answer: No. A router can simply forward DNS packets between the PC and DNS server.

Remember:

Router → Routing

DNS → Name resolution

Doubt — Can a Cisco router become a DNS server?

Answer: Yes.

```
ip dns server  
ip host PC1 192.168.0.101
```

But using a dedicated Windows/Linux DNS server is more common.

Doubt — What is `nslookup`?

Answer: A command used to query and troubleshoot DNS.

Doubt — What is the hosts file?

Answer: A local file containing static **hostname** → **IP** mappings.

Exam Points

- DNS = Domain Name System
- Purpose = Name resolution
- DNS port = 53
- A = IPv4
- AAAA = IPv6
- CNAME = Alias
- nslookup = DNS query/testing
- `ipconfig /displaydns` = DNS cache
- `ipconfig /flushdns` = Clear DNS cache
- `ip name-server` = Configure DNS server on router
- `ip dns server` = Make router a DNS server
- `ip host` = Static hostname/IP mapping
- `show hosts` = View router host table
- Routing gets the DNS packet to the DNS server; DNS resolves the name.



Command Review

Windows:

```
C:\Users\user>ipconfig /all  
C:\Users\user>nslookup name  
C:\Users\user>ipconfig /displaydns  
C:\Users\user>ipconfig /flushdns  
C:\Users\user>ping ip-address -n number
```

Cisco IOS:

```
R1(config)#ip dns server  
R1(config)#ip host hostname ip-address  
R1(config)#ip name-server ip-address  
R1(config)#ip domain lookup  
R1(config)#ip domain name domain-name  
R1#show hosts
```

